Sustainable Agriculture and Crop Water Planner

PROGRAMMING IN C | GROUP 8 | FIRST-YEAR B.TECH CSE

Group members: _____

Course / Section: _____ Submission date: _____

1. SDG Mapping

SDG 2 - Zero Hunger; SDG 12 - Responsible Consumption and Production.

2. Problem Statement

Small farms and gardens need to plan water use carefully. This console-based C program uses simplified, user-entered crop-water assumptions to compare estimated daily requirements with an available water budget.

3. Objectives

- Store crop names and assumed daily water rates.
- Estimate water requirement from cultivation area.
- Compare multiple crop records and calculate combined demand.
- Warn when estimated demand exceeds the available daily water budget.
- Generate a clear text-based planning summary.

4. Requirements and Data Units

Input	Unit / rule
Crop name	Single word, maximum 39 characters
Cultivation area	Square metres (m2); zero or positive
Assumed water rate	Litres per m2 per day; zero or positive
Available water budget	Litres per day; zero or positive

All crop rates are manually entered illustrative assumptions, not official crop coefficients or field measurements. A real planning tool would need local climate, soil, crop stage and irrigation-method data.

Program Design and Algorithm

5. Algorithm

1. Start the program and display the main menu.

2. Read a menu choice from 1 to 6.

3. If choice is 1, read crop name, area and rate; reject negative values; store the record in arrays.

4. If choice is 2, display all stored crop records.

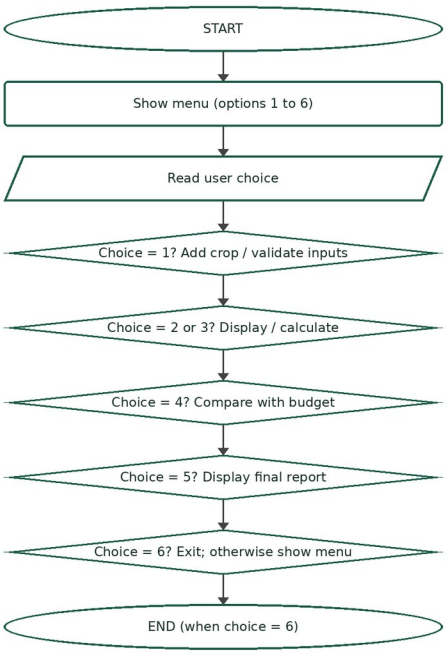
5. If choice is 3, calculate and display the sum of area multiplied by rate for all crops.

6. If choice is 4, read the daily budget and compare it with total required water; print a warning, balance or exact-match message.

7. If choice is 5, show a final report of all crops, total demand and the lowest assumed per-area rate.

8. If choice is 6, exit; otherwise return to the menu.

6. Flowchart



Implementation and Demonstration

7. C Programming Concepts

Concept	Application
Arrays	Store crop names, cultivation areas and rates
Functions	addCrop, calculateWater, totalWater, displayCrops, compareBudget, displayReport
Loops	Repeat menu and sum crop requirements
Switch-case	Handle menu choices
Conditions	Validate values and detect shortage
Strings	Store and display crop names

8. Formula and Worked Example

Estimated water (litres/day) = Area (m2) x Assumed rate (litres/m2/day).

Example: Wheat, 100 m2 at 4 L/m2/day = 400 L/day. Rice, 50 m2 at 6 L/m2/day = 300 L/day. Combined requirement = 700 L/day. With 650 L/day available, the estimated shortage is 50 L/day.

9. Compile and Run

Save the separate source file as Crop_Water_Planner.c. In a terminal with GCC installed, run:

gcc -std=c11 -Wall -Wextra Crop_Water_Planner.c -o crop_planner

./crop_planner (Linux/macOS) or crop_planner.exe (Windows)

10. Sample Input and Output

Menu choice 1: Wheat, area 100, rate 4. Menu choice 1 again: Rice, area 50, rate 6. Menu choice 4: water budget 650. Menu choice 5: final report.

Expected output: Wheat 400.00 L/day; Rice 300.00 L/day; total 700.00 L/day; WARNING: Water shortage = 50.00 L/day.

The complete commented C source code and a captured sample run are supplied as separate files.

Testing, Findings and Conclusion

11. Test Cases

Test	Input / condition	Expected result
1	Wheat: 100 m2, 4 L/m2/day; budget 500	400 L/day; 100 L remaining
2	Wheat: 100 m2, 4 L/m2/day; budget 350	400 L/day; shortage 50 L
3	Wheat: 100 m2, 4 L/m2/day; budget 400	Exactly matches budget
4	Area = -10	Reject negative area
5	Area = 0; rate = 4	0 L/day
6	Rate = -5	Reject negative rate
7	Wheat 400 L + Rice 300 L; budget 650	Total 700 L; shortage 50 L
8	Menu choice = 9	Invalid menu option message

12. Assumptions and Limitations

- Rates are supplied by the user and treated as fixed for the day; no agricultural source is claimed for the example rates.
- The same daily period and units are used for all crop records and the water budget.
- This is an educational estimation tool, not a sensor, rainfall forecast or irrigation recommendation.
- Records are kept in memory during one run; the maximum is 30 crops.
- Crop names are entered as single words; non-numeric entries in numeric fields may require restarting the program.

13. Future Improvements

Add file saving, support multi-word crop names, use verified crop-water datasets, and include rainfall, soil conditions, crop stage and irrigation efficiency.

14. Conclusion

The C program estimates daily water demand for several crops and compares the total with available water. It demonstrates arrays, functions, loops, switch-case, input validation and basic arithmetic while helping users understand water-budget planning. Results are estimates based on entered assumptions.

15. Source

Project_Workbook.pdf, Group 8 brief (page 10), common requirements (page 2), and assessment rubric (page 17). No external agronomic water-rate source was used.